

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 05 — BUILDING A SCADA HUMAN MACHINE INTERFACE

Student Edition

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Lab 05: Building a SCADA Human Machine Interface

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 4–5 hours

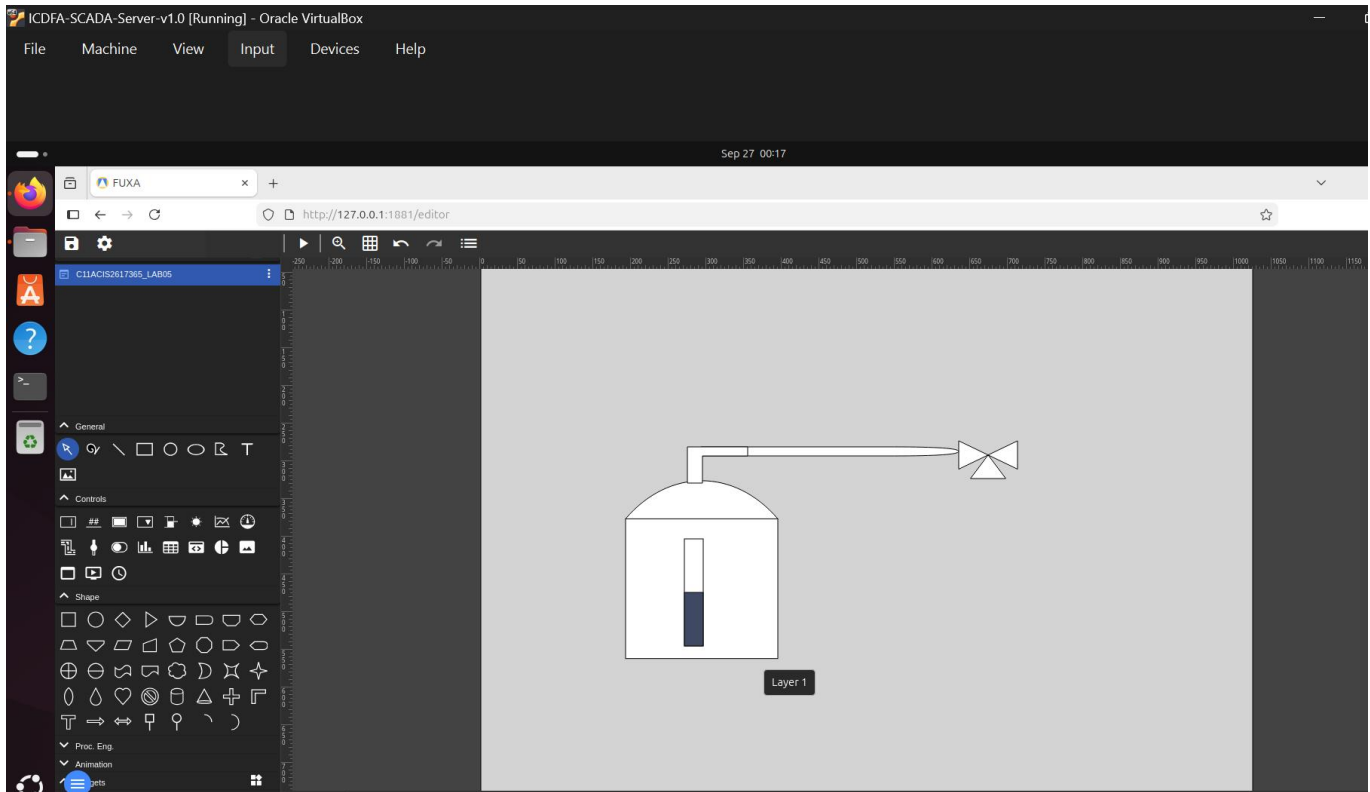
Submission deadline: As published in ICDFA E-Campus.

Procedure

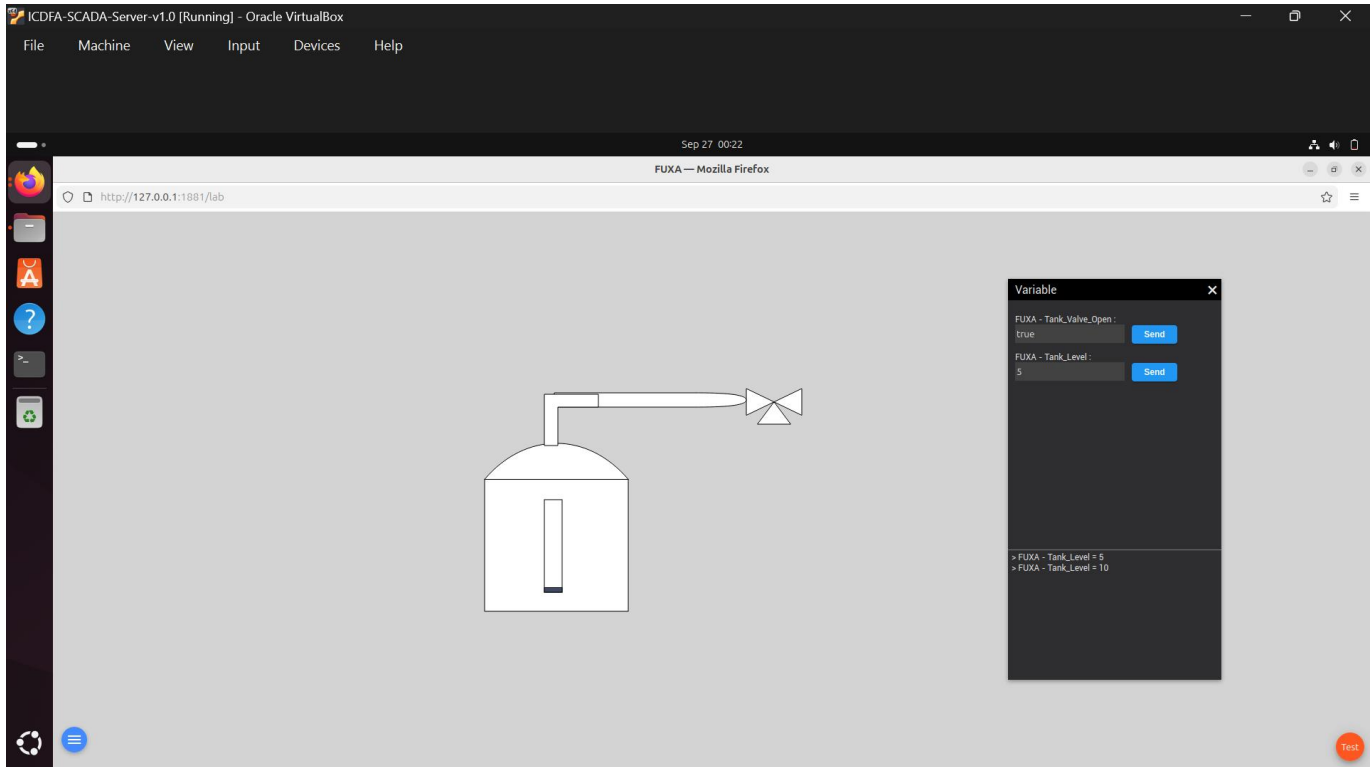
- Step 1.** Start the PLC and SCADA VMs in the isolated laboratory network. Start FUXA on the SCADA VM and open <http://127.0.0.1:1881/editor>.
- Step 2.** Create a New Project named with your registration number and LAB05, for example 2026XXXX_LAB05.
- Step 3.** Set a neutral gray background and enable the grid while arranging process graphics.
- Step 4.** From the Process Engineering objects, place a tank graphic. Add a bar gauge over/inside the tank and set its engineering range to 0–10 ft.
- Step 5.** Build a simple motor-actuated fill valve using the available valve/shape objects. Add a pipe from the valve to the tank. Keep the normal-state design visually simple.
- Step 6.** On the PLC VM, load/start the provided TankFillSim.st program in OpenPLC Runtime and confirm the status is Running.
- Step 7.** In FUXA, create or edit the Modbus TCP connection to the PLC at 192.168.1.10:502.
- Step 8.** Create/verify the Tank Level tag mapped to the OpenPLC input-register location used by the simulation (%IW80 / Input Register 80 in the teaching image).
- Step 9.** Create/verify the Tank Valve Open tag mapped to the OpenPLC discrete output used by the simulation (%QX0.0 / Coil 0 in the teaching image).
- Step 10.** Bind the bar gauge to the Tank Level tag. Configure Min = 0 and Max = 10.
- Step 11.** Bind the valve-state graphic to Tank Valve Open. Configure the normal/closed representation for value 0 and the open representation for value 1.
- Step 12.** Use a restrained high-performance style: neutral normal states, clear labels, and attention-drawing color only where it communicates a meaningful abnormal/alarm condition.
- Step 13.** Enable the PLC/FUXA connection and run the HMI. Observe the tank filling/draining and confirm the valve-state graphic changes with the process.
- Step 14.** Save/export the FUXA project and capture evidence of both the design view and running HMI.

Required Evidence & Submission

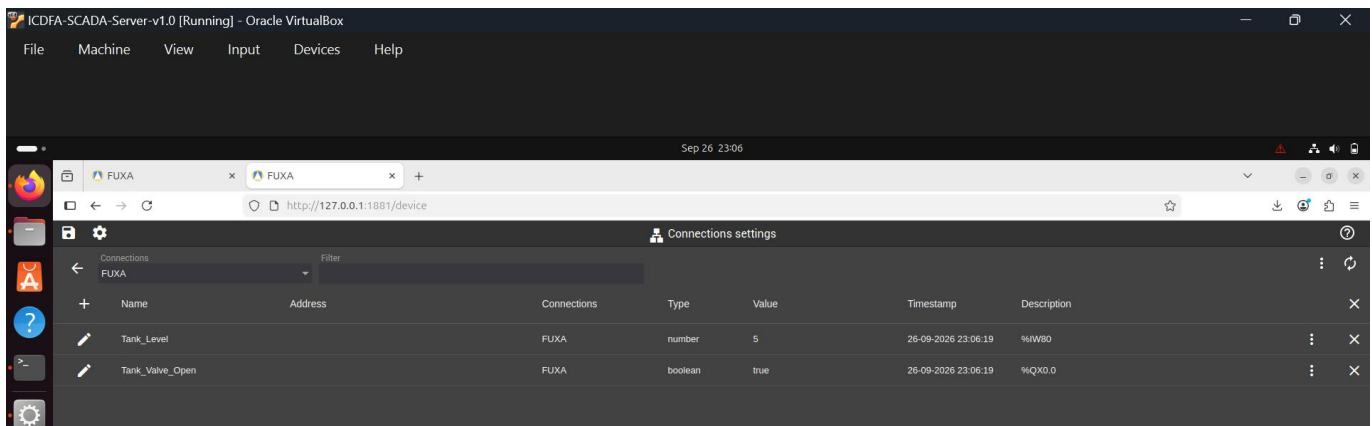
- Screenshot of the HMI in design/edit mode.



- Screenshot of the running HMI with a visible tank-level value.



- Screenshot of the configured PLC connection/tags with sensitive passwords excluded.



- **A short design rationale explaining how the interface supports situational awareness.**

Design Rationale (ANSI/ISA-101 Situational Awareness)

High-Performance HMI Rationale:

The human-machine interface was designed according to ANSI/ISA-101 high-performance standards to enhance operator situational awareness and mitigate visual fatigue. A neutral light-gray background reduces screen glare and eye strain during extended operational monitoring. Subdued monochromatic tones are applied to static equipment (vessels, piping, valve bodies) during normal state operation. Color is reserved strictly for dynamic variables and state indicators (such as liquid fill level and valve position status). Integrating a vertical bar gauge directly inside the tank vessel graphic provides an instant qualitative visual indicator of volume percentage and operational limits without forcing operators to manually read and interpret raw numerical values.

Knowledge Check

1. What is a P&ID and how does it influence HMI layout?

Response: A Piping and Instrumentation Diagram (P&ID) is a technical schematic showing plant equipment, vessels, piping routes, control valves, and sensor loops. It dictates the structural blueprint of the HMI layout, ensuring the graphic interface accurately reflects physical fluid paths and operational relationships.

2. Why do high-performance HMIs avoid unnecessary animation and excessive color?

Response: Excessive animation and vivid colors create visual clutter and operator fatigue. Muted palettes ensure normal operations remain visually quiet, allowing high-contrast primary colors (red, yellow, amber) to immediately command attention during abnormal conditions or process alarms.

3. How does the Tank Level tag differ from the Tank Valve Open tag in data type and behavior?

Response: The Tank Level tag is an analog data type (number / UInt mapped to Modbus register %IW80), representing continuous numerical values (0–10 ft). The Tank Valve Open tag is a discrete/digital data type (boolean mapped to Modbus coil %QX0.0), representing a binary state (0/false for Closed, 1/true for Open).

4. What could happen operationally if an HMI displays stale or incorrect PLC data?

Response: Displaying stale data severely compromises situational awareness, leading operators to make erroneous decisions. For instance, if an HMI incorrectly indicates a tank level is low when it is actually full, an operator might open supply valves, causing physical overflow, chemical spills, structural damage, or emergency shutdown.